

SEMESTER 2 EXAMINATIONS 2020/2021

MODULE: CA4023 - Natural Language Technologies

PROGRAMME(S):

DS BSc in Data Science
CAPT PhD-track

YEAR OF STUDY: 1,2,4

EXAMINER(S):

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Prof. Mathieu d'Aquin	(External)	External

TIME ALLOWED: 3 Hours

INSTRUCTIONS: Answer Question 5 and 3 other questions. All questions carry equal marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.

The use of programmable or text storing calculators is expressly forbidden.

Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

QUESTION 1**[TOTAL MARKS: 25]****Q 1(a)****[4 Marks]**

State two Document Classification tasks that can be tackled using Logistic Regression and two that can be tackled using Multinomial Logistic Regression.

Q 1(b)**[5 Marks]**

Logistic Regression is a **discriminative** approach whereas Naïve Bayes is a **generative** approach. Explain, using an example, the difference between the two approaches.

Q 1(c)**[8 Marks]**

Given the following training corpus

Document	Class
<i>a decent film</i>	positive
<i>very good film</i>	positive
<i>very bad movie</i>	negative
<i>not a good film</i>	negative
<i>not a bad movie</i>	positive

what label would a Naïve Bayes sentiment classifier which uses the words in a document as features assign to the following “document”?

really a very very bad film

Show the steps which the classifier follows to arrive at a prediction.

Q 1(d)**[8 Marks]**

Explain why the following sentence might represents a challenging test case for a sentiment classifier:

There is nothing original or creative and little enjoyable about this film.

What steps could be taken to help it to be classified accurately?

[End of Question 1]

QUESTION 2**[TOTAL MARKS: 25]****Q 2(a)****[9 Marks]**

Given the following training corpus

<s> a b </s>
<s> b b </s>
<s> b c </s>
<s> a a </s>

what probabilities will an unsmoothed bigram language model assign to the following sentences?

<s> c </s>
<s> b </s>
<s> b b </s>
<s> b a </s>
<s> a b c </s>

How will these probabilities change if add-one smoothing is employed?

Q 2(b)**[4 Marks]**

State one difference and one similarity between an n-gram language model and a language model trained using a feedforward neural network with a fixed context window.

Q 2(c)**[4 Marks]**

State one difference and one similarity between a language model trained using a feedforward neural network with a fixed context window and one trained using a recurrent neural network.

Q 2(d)**[8 Marks]**

State-of-the-art language models, such as GPT and BERT, are trained using transformer networks. Describe the architectural components of a transformer network that make these networks a good choice for modelling language.

[End of Question 2]

QUESTION 3

[TOTAL MARKS: 25]

Q 3(a)

[4 Marks]

Name three Natural Language Processing tasks that involve **sequence labelling**.
State one generative and one discriminative approach to sequence labelling.

Q 3(b)

[7 Marks]

Show the probabilities that result from training a Hidden Markov Model part-of-speech tagger using the following training corpus:

the D exam N paper N
check N book N
check V a D paper N
book V the D exam N

Assume that the label set contains the tags: *N (noun)*, *V (verb)*, *D (determiner)*

Q 3(c)

[7 Marks]

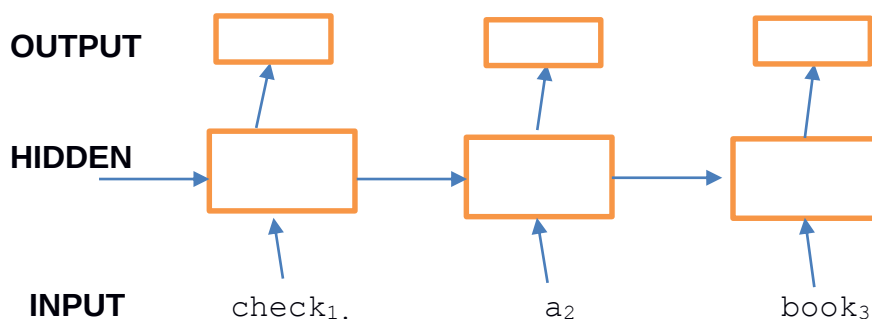
Assume that the model from Q3(b) is applied to the sequence *check a book* using Viterbi decoding. Show, step by step, how the decoding matrix is completed:

	check₁	a₂	book₃
N			
V			
D			

Q 3(d)

[7 Marks]

What kind of approach to sequence labelling is illustrated by the following diagram?



Explain what is happening at the output layer.

[End of Question 3]

QUESTION 4**[TOTAL MARKS: 25]****Q 4(a)****[8 Marks]**

Explain, using an example, what the linguist Firth meant when he said

You shall know a word by the company it keeps.

How can vectors be used to represent this idea?

Q 4(b)**[8 Marks]**

Given the following word co-occurrence matrix:

	brexit	date	film	family	twist
adventure	0	10	300	105	1
thriller	52	2	345	2	200
romcom	1	245	312	14	10

1. Show how the information in this matrix can be used to measure the similarity between the two words *adventure* and *thriller*.
2. Replace the counts in the above matrix with Positive Pointwise Mutual Information scores.

Q 4(c)**[9 Marks]**

What probability distribution is modelled by the Skipgram Negative Sampling *word2vec* classifier? Explain how this classifier is trained.

[End of Question 4]

QUESTION 5

[TOTAL MARKS: 25]

Q 5(a)

[15 Marks]

Describe what you consider to be the three most important ethical issues in the field of Natural Language Processing, and explain why you think these three issues are the most pressing.

Q 5(b)

[10 Marks]

For each of the issues identified in Q5(a), outline some steps that you think should be taken to mitigate the risks associated with these issues.

[End of Question 5]

[END OF EXAM]